

Chapter 23

Building AI Literacy and Competency in Social Work



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1 Introduction

Social workers worldwide live, work, and play in a digital and algorithmic era. This period is defined by the pervasive influence of digital technologies and algorithm-driven decision systems in nearly every aspect of life. The digital age began with the expansion of computers, mobile devices, and the Internet, transforming how people communicate, learn, and access services. The algorithmic age extends this transformation through the widespread use of data analytics, machine learning, and artificial intelligence (AI) to automate decisions, predict behavior, allocate resources, and deliver services. Together, these forces have reshaped how information is produced and shared, how systems are guided, how power and resources are allocated, and how human needs are understood and addressed. For the social work profession, that shift raises practical, ethical, and pedagogical questions about what we need to know and be able to do.

For social workers, this era brings both promise and peril, and the call to “harness technology for social good” has been highlighted as one of the grand challenges for social work. Data-driven, algorithmic tools and systems can improve access to care and training, identify unmet needs, augment learning, facilitate ease of communication, and enhance service coordination and quality. Yet these tools and systems can also perpetuate bias, obscure the sources of information and lines of accountability, and threaten privacy. To act ethically and effectively, social workers and their clients must gain AI literacy to understand how digital and AI systems function, whose values and assumptions are embedded in their design, and how to advocate for transparent, equitable, and human-centered uses of technology. AI literacy and the application of AI competencies have thus become essential for advancing social justice and protecting human dignity in a data-driven world, as well as for

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ensuring the social work profession's relevance and preparedness for the current era and the decades ahead.

AI literacy and the application of AI competencies are therefore essential for advancing social justice, protecting human dignity in the digital and algorithmic era, and ensuring the social work profession's relevance and preparedness now and into the future. Consequently, this chapter focuses on building capacity for AI literacy and competencies. Building on earlier chapters that document where and how AI is already used in practice, our aim here is to demonstrate how the profession can scale learning and workforce development, enabling social workers at every career stage to acquire the orientation and capabilities required for ethical and competent practice.

2 Defining AI Literacy and AI Competency in Social Work

2.1 Definitions: AI Literacy

2.1.1 Background: The Evolution of “Literacy”

The term “literacy” originally referred to the ability to read and write, express oneself, and communicate in writing (Freire, 2007). Over the past several decades, the notion of literacy has been extended to describe a set of abilities enabling individuals to engage meaningfully in specific domains: for example, digital literacy is defined as the capacity to use digital technology, communication tools, and networks; scientific literacy is defined as the ability to understand and engage in science-based reasoning (Laugksch, 2000); and data literacy refers to the ability to read, work with, analyze, and argue with data as part of inquiry (DiSalvo et al., 2014). Advances in technology, changes in work and society, and the proliferation of new forms of knowledge creation have driven this expansion of literacy concepts. Each of these literacies shares a common orientation: enabling expression, communication, and access to knowledge in a given domain.

2.1.2 AI Literacy: Concept and Framework

Under the broader umbrella of literacy, “AI literacy” has emerged as a distinct and critical domain. In essence, AI literacy builds on and extends prior literacies, such as digital and data literacy, by attending to the rapidly evolving characteristics of generative artificial intelligence (AI) technologies: their ability to learn from and act upon data, their autonomous behaviors, their integration into socio-technical systems, and their potential to reshape human tasks, decisions, and social systems.

In their influential work, Duri Long and Brian Magerko (2020) define AI literacy as “a set of competencies that enables individuals to evaluate AI technologies critically; communicate and collaborate effectively with AI; and use AI as a tool online,

at home, and in the workplace” (Long & Magerko, 2020). Their framework proposed five thematic dimensions: (1) understanding AI concepts and types, (2) recognizing AI’s capabilities and limitations, (3) comprehending underlying AI processes, (4) addressing ethical and societal implications, and (5) examining public perceptions of AI (Long & Magerko, 2020). This work situates AI literacy beyond mere technical or programming knowledge; it emphasizes the capacity for critical engagement, responsible use, and effective human–AI interaction.

Subsequent scholarship has refined this conceptualization. In search of a sound theoretical foundation to define, teach, and evaluate AI literacy, a recent exploratory review was conducted to conceptualize the newly emerging concept “AI literacy.” Grounded in a review of 30 existing peer-reviewed articles, this review identifies four core aspects of AI literacy: knowledge/understanding, use/interaction, evaluation, and ethics/reflective awareness (Tsz et al., 2021). Further empirical work situates AI literacy as a multidimensional construct that encompasses effective human-AI collaboration, creative idea generation, content evaluation, and step-by-step collaboration (Liu et al., 2025; Perchik et al., 2023). Thus, the term “literacy” in this context retains its classical grounding while being extended to the domain of AI technologies.

2.2 Definition of AI Competencies

2.2.1 Background: Competency in Professional Education

Across different professions, competence is widely defined as the habitual and judicious integration of knowledge, skills, communication, reasoning, values, and reflection in real settings for the benefit of those served (Epstein & Hundert, 2002). This definition emphasizes that competence is not only what a practitioner knows but how that knowledge is enacted responsibly in context. Assessment frameworks further distinguish levels of performance and competence that reflect knowing, knowing how to do something, the ability to show others how, and doing and accomplishing the critical tasks of the domain; thus, these levels underscore that competence is observable in practice and must go beyond factual understanding (Miller, 1990a). Contemporary competency-based education extends these ideas, portraying competencies as multidimensional, developmental, and context-contingent outcomes that organize curricula and accountability (Frank et al., 2010).

2.2.2 AI Competencies: Scope and State of the Literature

Translating these principles, AI competencies refer to the demonstrable ability to apply AI-related knowledge, skills, and values to concrete tasks, such as selecting, configuring, deploying, supervising, and evaluating AI systems, in alignment with professional standards and societal obligations. In education, researchers discuss

concepts similar to AI literacy, but with a slightly narrower focus than the above definitions. They distinguish between knowing how AI works and being able to use it responsibly in real-world situations, emphasizing the importance of ethics, evaluation, and keeping people at the center of AI use (Chiu, 2025; Chiu et al., 2024). International policy and standards entities have begun to codify AI competencies for specific audiences. For example, UNESCO's frameworks for students and teachers specify domains such as a human-centered mindset, ethics of AI, AI foundations and applications, AI pedagogy, and professional learning, each with progression levels ranging from acquisition to creation (UNESCO, 2024, 2025). Beyond education, organizational research conceptualizes AI competence as the capability to integrate AI technologies with complementary resources and skills to achieve optimal performance, thereby highlighting that competencies can be defined at both individual and institutional levels (Mikalef et al., 2023).

2.3 AI Literacy and AI Competencies in the Social Work Context: Specificities and Distinctions

General AI literacy frameworks establish a necessary baseline, but the discipline and profession of social work demand tailored adaptations of general AI literacy and competencies frameworks, given the profession's core mission, values, and varied contexts of practice and research. More general approaches define AI literacy as the capacity to understand AI concepts and types, recognize capabilities and limits, grasp underlying processes, and consider societal implications (Liu et al., 2025; Long & Magerko, 2020; Perchik et al., 2023; Tsz et al., 2021); and those components still apply, yet their weighting and practical expression shift in a profession grounded in person-centered, values-driven, context-sensitive practice.

Social work is fundamentally relationship-centered, and so AI literacy and AI competencies in the context of social work must begin with the effects of digital and algorithmic tools on trust, power, and rapport in the worker–client relationship. Research on digitalization in social work documents both benefits and strains for relational practice, making critical reflection on human–AI mediation a core literacy element for social work, rather than an optional add-on (Nordesjö et al., 2022). An AI-literate social worker should be able to anticipate when technology reconfigures attention, authority, and the meaning of presence and human connection in direct practice with individuals and families.

Social workers routinely steward highly sensitive information; therefore, AI literacy and AI competencies in this field must prioritize data flows, consent, retention, secondary use, and the limitations of de-identification, alongside evolving professional standards. Ethics scholarship in social work highlights boundary management, confidentiality, and documentation as recurring challenges intensified by digital tools, and the profession's technology standards set expectations that now extend to algorithmic systems (Reamer, 2013). Practitioners, therefore, need

operational literacy in how these standards apply when AI is embedded in assessment, referral, or service delivery.

Many social work agencies operate under tight budgets and uneven IT support; therefore, AI literacy must incorporate practical knowledge about organizational readiness, governance, and maintenance under constrained conditions. Reviews of information and communication technologies used in social work show that without investment in training, supervision, and evaluation, adoption is fragile and uneven (Chan & Holosko, 2016; Perron et al., 2010); literacy thus includes the ability to appraise feasibility, advocate for safeguards, and plan for responsible implementation. This organizational lens transforms generic AI knowledge into field-ready literacy, attuned to the realities of staffing, infrastructure, and accountability.

Because social workers work with clients who are vulnerable to experiencing digital exclusion, AI literacy in social work must be equity-attuned and accessibility-aware. Decades of scholarship on digital inequalities demonstrate how access, skills, and usage patterns track existing social disadvantages. In social work services, this means asking who benefits, who is burdened, and how to redesign or supplement tools to address language access, connectivity, disability accommodations, and cultural fit. Equity, therefore, becomes a baseline condition for AI literacy and AI competencies rather than a downstream consideration.

2.4 Working Definitions for This Chapter

For this chapter, we therefore define AI literacy in social work as the integrated knowledge, interpretive judgment, and ethical orientation that enable social work students, practitioners, educators, and researchers to (1) explain, in accessible terms, how AI systems used in human services are designed, trained, validated, and governed; (2) appraise capabilities, limitations, error modes, and fairness implications in context; (3) communicate transparently with clients and interprofessional teams about whether and how AI is used, preserving rapport, client autonomy, and culturally responsive practice; (4) safeguard confidentiality, informed consent, and prudent data stewardship across data life cycles; and (5) anticipate and mitigate inequities in access and outcomes for digitally marginalized populations. This definition extends general AI literacy scholarship by centering on relational practice, ethical research conduct, professional standards, organizational constraints, and digital inclusion as constitutive conditions of literate AI engagement in social work.

AI competencies in social work refer to the enacted, context-appropriate abilities that professionals operationalize AI literacy in practice, teaching, administration, and research, considering organizational, regulatory, and equity constraints. Competent performance is evidenced by the capacity to: (1) select, configure, and integrate AI tools into micro, mezzo, and macro workflows, curricula, and study designs in alignment with agency policy, law, and institutional review requirements; (2) maintain ongoing oversight of system performance, make necessary adjustments, and document procedures using model and dataset records, while auditing

for distribution drift, errors, and unequal impacts across groups; (3) protect privacy and consent across data collection, sharing, retention, linkage, and secondary use in service delivery, classrooms, and research; (4) sustain relational integrity and support the autonomy of clients, students, and research participants when digital systems mediate contact; (5) ensure accessibility and inclusion for digitally marginalized populations in services, learning environments, and studies; and (6) participate in governance and accountability, including algorithmic impact assessment, incident response, documentation for reproducibility, and continuous improvement. This performance-focused formulation aligns with competency theory, which emphasizes the observable integration of knowledge, skills, values, and judgment. It also reflects emerging AI competency frameworks that prioritize a human-centered, ethical application and evaluation within the CSWE's outcomes-based approach to professional practice and research.

2.5 Differences and Connections Between AI Literacy and AI Competency in Social Work

AI literacy and AI competency in social work are analytically distinct yet developmentally linked for students, practitioners, educators, and researchers in the field (Table 23.1). Literacy is broad, conceptual, and values attuned. It equips these groups to explain how AI used in human services is designed and governed, to judge capabilities and limits, to communicate use transparently in practice, teaching, and research, and to anticipate equity and privacy risks (Long & Magerko, 2020). AI competency is role-specific and performance-based. This is evidenced by their ability to select and integrate tools into agency workflows, curricula, and study designs; monitor and document impacts, protect confidentiality and consent; sustain relational practice with clients and participants; ensure accessibility; and participate in governance and continuous improvement. In short, literacy orients judgment, while competence demonstrates that judgment operates within organizational, legal, and equitable constraints.

The relationship between literacy and competency is sequential, reciprocal, and situated within the ethical and practical realities of social work (Table 23.1). Literacy is a prerequisite and compass for competent action, steering client-centered, culturally responsive, and human rights-oriented choices in services, classrooms, and studies. Competent performance, in turn, refines literacy by generating situated insights about model behavior, documentation gaps, workflow fit, and unintended effects on rapport and access, which then inform critical understanding, pedagogy, and policy advocacy. The practical implication is a literacy-to-competency pathway aligned with competency-based social work professional education. Programs and agencies should build shared literacy foundations in concepts, ethical reasoning, data stewardship, and equity analysis, then assess competency through observable performance in authentic tasks that develop from understanding core concepts, to being able to explain and design appropriate actions, and finally to confidently and independently performing those actions in classroom, field, and research settings

Table 23.1 AI literacy versus AI competency concepts in social work

Component	AI literacy	AI competency
Definition	Integrated knowledge, interpretive judgment, and ethical orientation that enable social workers to explain how AI systems used in human services are designed, trained, validated, and governed; to appraise capabilities, limitations, error modes, and fairness implications in context; to communicate transparently with clients and teams while preserving rapport and client autonomy; to safeguard confidentiality and informed consent across the data life cycle; and to anticipate and mitigate inequities in access and outcomes	Enacted, context-appropriate abilities through which professionals operationalize AI literacy under organizational, regulatory, and equity constraints. Evidenced by the capacity to select, configure, and integrate tools within workflows; monitor, calibrate, and document system behavior and impact; protect privacy and consent; sustain relational integrity and client self-determination when technology mediates contact; ensure accessibility for digitally marginalized groups; and participate in governance, audit, and continuous improvement
Scope	Broad, conceptual, and value based. Oriented toward understanding, critique, and responsible use	Role-specific and performance-based. Oriented toward effective implementation, evaluation, and innovation
Purpose	To cultivate informed, ethical, and reflective practitioners who can judge the appropriateness of AI, communicate its use transparently, and lay a foundation for safe and equitable adoption aligned with social work values	To develop competent Social Work practitioners, educators, and researchers who can ethically apply AI tools to enhance practice, learning, and knowledge generation

(Council on Social Work Education, 2022; Miller, 1990b). In this model, the purpose of literacy is to enable informed, ethical, and reflective judgment about if and how AI should be used, while the purpose of competency is to ensure that such judgment is enacted safely and effectively to enhance practice quality, mitigate bias, and promote equity.

3 Aligning AI Use and Training with NASW Code of Ethics and CSWE Competencies

3.1 National Association of Social Workers (NASW) Code of Ethics

The NASW Code of Ethics is a “set of standards that guide the professional conduct of social workers” (NASW, 2021). It outlines six core values that can guide social workers’ engagement with AI and algorithmic systems: service, social justice, the dignity and worth of the person, the importance of human relationships, integrity, and competence. These values prompt social workers to examine the

ethical implications of AI, including privacy, systemic bias, and transparency, and to advocate for the equitable and responsible use of AI. From a social justice perspective, social workers must consider how AI may impact access to services, resource allocation, and ensure that human rights are upheld. The value of dignity and worth of the person invites practitioners to monitor algorithms for unintended harm and to promote client agency in digital environments. Emphasizing the importance of human relationships reminds social workers to use AI to enhance, not replace, empathy and human connection. Competent and ethical practice in the algorithmic era requires AI literacy to evaluate AI tools and their impact on client well-being, dignity, and self-determination. Guided by these core professional values, social workers are uniquely positioned to collaborate with data scientists and AI developers, ensuring that emerging technologies serve clients, honor human relationships, and promote social justice. The Code of Ethics provides both purpose and direction for advancing AI literacy and competency across social work practice, education, and research. The CSWE competencies further translate these values into the specific knowledge, skills, and behaviors needed to practice effectively in an AI-driven environment.

3.2 CSWE Policy and Accreditation (EPAS) Standards Competencies and AI

The CSWE competency model specifies the knowledge and skills expected of graduating social workers (Council on Social Work Education, 2022). In the algorithmic era, the central task for social work educators and accredited programs is to operationalize those competencies within settings where assessment, service delivery, policy, and evaluation are increasingly mediated by AI (Rodriguez et al., 2024). Research suggests that AI literacy should be integrated across core social work competencies, rather than being treated as an elective or a single stand-alone course, and that curricular integration should align with existing accreditation expectations and professional values (Ahn et al., 2025; Rodriguez et al., 2024). Aligning instruction on AI with accreditation standards and risk-sensitive pedagogies can help ensure that graduates are prepared to manage the ethical, equity, and accountability implications of algorithmic systems (Ahn et al., 2025). Consequently, CSWE's vision of competent, ethical, evidence-informed practice should be enacted in the very contexts where algorithmic systems already shape decisions and opportunities (Council on Social Work Education, 2022).

This subsection proceeds in two parts. We first illustrate how each of the nine CSWE EPAS competencies could be updated or enhanced in response to AI's current transformation of practice. Second, we introduce components of a new AI competency for consideration as revisions to the 2029 EPAS are underway.

What follows are suggested contexts and practice behaviors for each of the nine competencies relevant to AI in social work practice (summarized in Table 23.2). AI Literacy would be incorporated into Generalist practice behaviors,

Table 23.2 Integrative framework: AI literacy and AI competence across CSWE nine competencies and NASW Code of Ethics

CSWE EPAS 2022 Competency	Relevant NASW ethical principle	AI literacy (awareness, critical reflection)	AI competence (applied skills, tool mastery, evaluation)
1. Demonstrate ethical and professional behavior	Integrity, competence, service	Understands ethical implications of AI (privacy, bias, transparency); recognizes limits of algorithmic approaches and need for human oversight	Applies ethical reasoning to select, use, and monitor AI tools; maintains transparency, confidentiality, and professional integrity in digital contexts
2. Advance human rights and social, racial, economic, and environmental justice	Social justice, service, political action	Understands how AI affects distribution of resources, surveillance, and social control	Engages in advocacy and policy reform to ensure AI advances human rights and digital equity
3. Engage anti-racism, diversity, equity, and inclusion in practice	Dignity and worth, social justice	Recognizes digital inequities and algorithmic bias as new forms of systemic oppression	Uses and evaluates AI systems to promote inclusion and equity; advocates for bias mitigation and fair access to technology
4. Engage in practice-informed research and research-Informed practice	Competence, integrity, scientific inquiry	Critically interprets AI-based research, data provenance, and potential bias	Uses AI ethically for data analysis, modeling, or evaluation; ensures transparency and validity in AI-driven research
5. Engage in policy practice	Social justice, service, political action	Understands policy implications of AI (data governance, regulation, digital rights, privacy)	Participates in policy analysis and advocacy for equitable and ethical AI use in human services; monitors and assesses equity when systems use AI to identify human needs and allocate resources
6. Engage with individuals, families, groups, organizations, and communities	Dignity and worth, human relationships, service	Recognizes relational and cultural impacts of AI-mediated engagement; reflects on trust and authenticity in digital interactions	Uses AI-enhanced tools (e.g., case management platforms, chatbots) to augment but not replace human empathy and connection
7. Assess individuals, families, groups, organizations, and communities	Integrity, competence, privacy and confidentiality	Understands data-driven assessment tools and their potential biases	Integrates AI-generated insights with professional judgment; ensures validity and fairness in assessment

(continued)

Table 23.2 (continued)

CSWE EPAS 2022 Competency	Relevant NASW ethical principle	AI literacy (awareness, critical reflection)	AI competence (applied skills, tool mastery, evaluation)
8. Intervene with individuals, families, groups, organizations, and communities	Service, dignity and worth, competence	Understands potential benefits and risks of AI-supported interventions	Applies AI tools (e.g., digital therapeutics, predictive supports) ethically, monitors for unintended harm, and centers client agency
9. Evaluate practice with individuals, families, groups, organizations, and communities	Competence, integrity, accountability	Understands how AI informs evaluation metrics and data ethics	Uses AI-driven analytics responsibly to evaluate practice outcomes; collaborates with data experts to ensure ethical use

Note. CSWE competency language and NASW ethical principles are drawn from CSWE (2022) and NASW (2021), respectively. Descriptions of AI literacy and AI competence are author-developed syntheses informed by these documents and the broader interdisciplinary literature on artificial intelligence and professional practice.

and AI Competence into Advanced areas of specialized practice. These contexts and practice behaviors serve as a starting point that reflects knowledge, skills, values, and/or cognitive and affective processes relevant to AI in social work practice.

3.2.1 Competency 1: Demonstrate Ethical and Professional Behavior

AI literacy begins with understanding the ethical implications of algorithmic systems, including privacy, data security, transparency, and bias. Social workers must recognize the limits of automated decision-making and the critical need for human oversight. AI competence involves applying ethical reasoning to the selection, use, and monitoring of AI tools; maintaining professional integrity, confidentiality, and transparency in digital contexts; and ensuring that technology aligns with the NASW Code of Ethics.

3.2.2 Competency 2: Advance Human Rights and Social, Racial, Economic, and Environmental Justice

Social workers with AI literacy understand how algorithmic systems affect the distribution of resources, surveillance practices, and mechanisms of social control. They critically assess whether these tools advance or undermine human rights and social equity. AI-competent social workers engage in advocacy, policy reform, and public education to ensure that AI technologies are used to promote, rather than restrict, rights and opportunities.

3.2.3 Competency 3: Engage Anti-racism, Diversity, Equity, and Inclusion in Practice

AI literacy includes awareness of digital inequities and algorithmic bias as potentially new forms of systemic oppression. Social workers must learn to identify how data and design choices reproduce historical injustices. Building competence involves using and evaluating AI systems to promote inclusion, transparency, and fairness. Practitioners should advocate for mitigating bias, equitable data practices, and fair access to technology that benefits all communities.

3.2.4 Competency 4: Engage in Practice-Informed Research and Research-Informed Practice

An AI-literate social worker critically interprets AI-generated research, understanding issues of data provenance, sampling bias, and algorithmic validity. Competence involves using AI ethically in data analysis, modeling, and evaluation while ensuring that results are grounded in sound social science methods. Social workers contribute to AI-driven research that reflects human values, community priorities, and social good.

3.2.5 Competency 5: Engage in Policy Practice

AI literacy encompasses an understanding of the policy implications of AI, including data governance, regulation, digital rights, and privacy. Social workers must be able to analyze how policy frameworks shape the use of AI in human services. AI competence involves participating in advocacy and policy design to promote equitable and ethical AI use, monitoring the effects of automated systems on service needs identification, and ensuring that AI supports fairness in resource allocation.

3.2.6 Competency 6: Engage with Individuals, Families, Groups, Organizations, and Communities

AI literacy entails awareness of the relational and cultural dimensions of technology-mediated engagement. Social workers reflect on how digital tools affect trust, authenticity, and power in human relationships. Competence means skillfully using AI-enhanced platforms, such as case management systems, chatbots, or telehealth supports, to augment rather than replace human empathy and connection. Technology should complement rather than replace authentic relational practice.

3.2.7 Competency 7: Assess Individuals, Families, Groups, Organizations, and Communities

An AI-literate practitioner understands the structure, purpose, and potential biases of data-driven assessment tools. They recognize when algorithmic recommendations may reinforce inequities or obscure contextual factors. Competence involves integrating AI-generated insights with professional judgment, maintaining fairness and validity in assessment, and ensuring that data use aligns with ethical standards for privacy and confidentiality.

3.2.8 Competency 8: Intervene with Individuals, Families, Groups, Organizations, and Communities

AI literacy includes understanding both the potential benefits and the risks of AI-supported interventions, such as predictive analytics, digital therapeutics, or automated support tools. Competence means applying these tools ethically and responsibly, continuously monitoring for unintended harm, and centering client agency and informed consent. Social workers must ensure that interventions guided by AI are transparent, accountable, and person-centered.

3.2.9 Competency 9: Evaluate Practice with Individuals, Families, Groups, Organizations, and Communities

AI literacy in evaluation involves understanding how AI informs outcome measurement, program analytics, and data ethics. Competent practitioners utilize AI-driven analytical tools to evaluate practice outcomes responsibly and collaborate with data experts to ensure the ethical use of client information. By integrating AI thoughtfully into evaluation processes, social workers can enhance evidence-based practice while upholding values of accountability and integrity.

3.3 *Proposing a New Competency for the 2029 EPAS: “Engage AI and Algorithmic Systems Informed Practice”*

While AI literacy should be infused throughout the existing competencies (Boetto, 2025; Goldkind, 2021), the pace and scope of technological transformation suggest the need for an explicit competency focused on AI and digital ethics. This new competency would articulate expectations for social workers to understand, evaluate, and apply AI technologies in ways that advance human well-being, promote social justice, and preserve the profession’s relational and ethical core.

The Council on Social Work Education (CSWE) is currently updating the Nine Competencies outlined in the 2022 Education and Policy Accreditation Standards (EPAS) (CSWE, 2022). Given the likelihood that AI and the algorithmic era will transform nearly every aspect of social work practice and the practice context in the coming decade, some scholars in the social work profession have been proposing a tenth competency focused on AI (Rodriguez et al., 2024). The tenth competency could be called “Engage AI and Algorithmic Systems Informed Practice.”

Under the CSWE EPAS 2022, each of the nine competencies is followed by a set of Practice Behaviors at the Generalist level for all programs. Then, each program develops its own set of Practice Behaviors at the Advanced level for each area of specialized practice it offers. For the proposed tenth competency, we recommend that AI literacy be incorporated into the Generalist practice behaviors and that more advanced applications of AI be developed within a program’s areas of specialization. This would ensure basic knowledge at the Generalist level and encourage deeper engagement in areas of specialization. In the proposed three-tier system of AI literacy and competency outlined below, Generalist practice behaviors align with Tier I, which encompasses foundational AI literacy knowledge. In contrast, advanced practice behaviors would reflect Tier II (intermediate) or Tier III (advanced) level AI knowledge, skills, values, and cognitive/affective processes. The levels of AI competency and corresponding training recommendations are described below.

4 Building Capacity for Social Work AI Readiness: A Three-Tier Approach

Modern social work must build capacity for AI readiness through deliberate professional development spanning all levels of practice, research, education, policy, and ethics. As discussed in earlier sections, emerging ethical standards and accreditation guidelines are beginning to acknowledge technology and AI as integral to competent professional formation (Hodgson et al., 2022; Nuwasiima et al., 2024). The Council on Social Work Education (CSWE) has identified increasing knowledge of AI and its implications as a strategic priority for social work programs (Council on Social Work Education, 2025). Likewise, leading scholars argue that social workers in all roles will need at least basic AI literacy, the ability to understand and critically evaluate AI systems, to uphold social justice and effectively advocate for clients in an AI-driven society (Ahn et al., 2025; BASW, 2025; Liu et al., 2025; Rodriguez et al., 2024). Given this broad and cross-cutting impact, it is imperative to prepare the entire workforce, students, practitioners, faculty, and agency leaders to engage with AI ethically and competently. Professional development (PD) for AI readiness thus encompasses both AI literacy (a foundational understanding for all) and AI competency (applied skills for practice) in social work.

This section proposes a three-tier PD framework to scaffold AI readiness, illustrated in Table 23.3: Tier 1 for foundational AI literacy among all social work

Table 23.3 Three-tier approach to building AI literacy and AI competencies in social work

Tier	Level	Audience	Focus of learning	Key learning goals/Outcomes	Examples of application
I	Foundational: Foundational AI Literacy for All Social Workers	All social work students, faculty, field instructors, agency staff, and supervisors	Awareness and basic understanding of AI concepts, benefits, risks, and ethical issues	• Recognize everyday AI applications in professional and personal contexts. • Understand privacy, consent, and data safeguards. • Identify potential bias and ethical implications. • Cultivate curiosity and critical thinking about AI's role in human services.	Recognizing AI features in tools like documentation software, scheduling apps, or case management platforms; following ethical and privacy guidelines in digital communication
II	Intermediate: Applied AI Competencies for Practitioners, Educators, and Researchers	Social work practitioners, educators, researchers, administrators, and policy professionals seeking to integrate AI into their work	Application of AI tools to enhance professional functions and outcomes	• Develop and apply AI-informed tools ethically in service delivery, teaching, or research. • Analyze data responsibly and identify bias. • Use AI to improve efficiency and outcomes while upholding social work values.	Using AI analytics in program evaluation; employing chatbots for resource navigation; integrating AI-assisted writing or learning tools into teaching; employing predictive analytics for policy or community assessment

(continued)

Table 23.3 (continued)

Tier	Level	Audience	Focus of learning	Key learning goals/Outcomes	Examples of application
III	Advanced: Advanced Leadership and Innovation in AI for Social Good	Social work leaders, scholars, researchers, and interdisciplinary innovators	Leadership, innovation, and governance in AI development, policy, and ethical oversight	• Lead interdisciplinary AI projects- Shape ethical AI policies and regulatory practices. • Advocate for human-centered and socially just AI systems. • Conduct or guide AI-related research and innovation.	Developing ethical AI governance policies, contributing to interdisciplinary AI research, designing AI systems that reflect social work’s core values and justice-oriented frameworks

professionals; Tier 2 for intermediate training focused on the applied use of AI in practice and administration; and Tier 3 for advanced leadership and governance capacities. This tiered approach aligns with recommendations by social work education experts who outline the need for foundational digital skills, practice-integration competencies, and technology leadership across the profession. By building capacity at these three levels, the social work profession can proactively meet emerging standards and ensure that AI integration aligns with core values and ethical commitments (Reamer, 2023).

4.1 Tier 1: Foundational AI Literacy for All Social Workers

Tier 1 focuses on broad foundational AI literacy for the widest audience (see Table 23.3). The audience at this tier includes social work students, educators, researchers, and practitioners across settings who are new to AI concepts. The focus of PD at this tier is basic awareness and conceptual understanding of AI: participants learn fundamental terminology, how AI is being used in society and social services, and core ethical issues. At this foundational level of AI literacy, the aim is to build comfort and dispel myths, ensuring learners understand what AI is and is not capable of. Key learning goals include recognizing common AI applications in social work, understanding the potential benefits and risks of these tools, and developing an initial critical perspective grounded in social work values and ethics. By the end of Tier 1 training, individuals should be able to engage in informed conversations about AI and identify situations in practice or policy where AI might be relevant. As Table 23.3 indicates, Tier 1 is about establishing a baseline upon which

more advanced competencies can be built, ensuring that the learning remains accessible and broadly applicable.

Examples of applications at this tier include introductory and low-stakes ones. For instance, a social work program might incorporate a brief AI literacy module into an orientation course, familiarizing all incoming students with AI terminology and ethical dilemmas. Faculty and staff could attend general workshops or lunch-and-learn sessions on “AI essentials” that introduce tools like ChatGPT and discuss their appropriate use in academic or practice settings. In agency contexts, Tier 1 may involve basic training for practitioners on how AI-driven systems work, accompanied by guided discussions on client consent and bias. The goal is not to produce experts, but to ensure everyone in the social work ecosystem shares a common understanding of AI basics. This broad foundation creates a shared language and awareness, which is crucial for scaling up more specialized AI learning in the following tiers.

4.2 Tier 2: Intermediate AI Competencies in Practice, Education, and Research

Building on this foundation, Tier 2, “Applied AI Competencies for Practitioners, Educators, and Researchers” shifts the focus from general literacy to the practical application of AI in specialized social work roles. At this intermediate level, the focus of learning centers on applying AI tools to enhance professional functions and outcomes (see Table 23.3).

In alignment with the six core components of AI competency outlined earlier, Tier 2 training emphasizes the development of skills and ethical practice in the use of AI in real-world social work contexts. Social workers in this tier learn to develop, adapt, and apply AI-informed tools in their specific practice or educational environments, whether in direct service delivery, program administration, teaching, or research, ensuring they are employed in an ethically sound manner that upholds social work’s core values. A key learning outcome is the ability to critically analyze data and AI-generated outputs responsibly, which includes identifying biases in data or algorithms and interpreting automated insights with an understanding of context and limitations. Practitioners are also taught to leverage AI to improve the efficiency and effectiveness of interventions or administrative processes, while maintaining a focus on human well-being and social justice.

Participants who complete Tier 2 can translate foundational knowledge into competent action. They not only grasp AI concepts but can also make informed decisions about when and how to use AI in practice, selecting appropriate tools and approaches to solve problems. For instance, a social work program evaluator at this stage might use AI-driven analytics to assess the outcomes of a service program, thereby enhancing program evaluation with data-driven insights that can inform improvements. Likewise, a practitioner could employ a chatbot or virtual assistant

to help clients navigate resources and services, provided that the tool is accessible, culturally sensitive, and maintains client confidentiality. In an educational setting, instructors can integrate AI-assisted writing or learning tools to support student learning. At the same time, administrators can use predictive analytics to identify community needs and proactively allocate resources for preventive programs. Taken together, these examples (see Table 23.3) demonstrate how Tier 2 solidifies an intermediate level of AI competence that builds directly upon the Tier 1 literacy foundation, effectively bridging the gap between knowing about AI and adeptly using AI to enhance social work practice.

4.3 Tier 3: Advanced AI Leadership and Innovation

Tier 3, “Advanced Leadership and Innovation in AI for Social Good,” represents the highest level of professional development in this model and targets those in leadership and pioneering roles. This tier is designed for social work leaders, senior administrators, experienced scholars, and interdisciplinary innovators who will guide the strategic integration of AI in social services and advocate for the ethical use of technology at organizational and policy levels. The focus of learning at Tier 3 is on leadership, innovation, and governance in AI development, policy, and ethical oversight (see Table 23.3).

Tier 3 training cultivates the capacity to lead and manage AI-driven initiatives.

Learners also strengthen the ability to shape ethical AI policies and regulatory practices, positioning themselves as influential voices for human rights and social justice in broader technology governance conversations. Consistent with the highest-level AI competencies previously outlined, Tier 3 emphasizes competencies related to policy advocacy, organizational leadership, and innovation management in the context of AI. Professionals at this level are prepared to champion human-centered and socially just AI systems, ensuring that any AI adoption within their agencies or communities is aligned with core social work values and actively counteracts biases or inequities. Moreover, Tier 3 participants are equipped to conduct or guide AI-related research and development, allowing them to contribute to the evidence base on AI in social work and to supervise interdisciplinary teams in creating new AI tools or approaches grounded in the profession’s ethical principles and commitment to social good.

Examples of application at the Tier 3 level (Table 23.3) illustrate the transformative leadership roles envisioned for this stage. For instance, a social work executive or policy leader might lead the development of comprehensive AI governance guidelines for a human services organization or contribute to government regulations that ensure AI tools are used responsibly and equitably in the social sector. A scholar at this level could collaborate with computer scientists to design an AI system that addresses a community issue, such as an algorithm to better target social services to those most in need, while embedding transparency and fairness into its

design from the outset. Others may participate in interdisciplinary research initiatives to assess the societal impacts of AI or to develop new AI-driven interventions for the greater good, rigorously examining outcomes and their ethical implications. Through Tier 3, the model produces social work professionals who are not only capable of navigating and utilizing AI in practice but also lead its ethical integration and drive innovation. In doing so, these leaders ensure that the evolution of AI in social work remains grounded in human rights and social justice principles.

4.4 Cross-Tier Regulatory and Ethical Considerations

Across all three tiers of AI readiness, certain regulatory and ethical considerations must be interwoven to ensure compliance with laws and alignment with social work values. Regardless of skill level, social workers using AI are still bound by fundamental regulations, such as applicable privacy rules under the Health Insurance Portability and Accountability Act (HIPAA), the Family Educational Rights and Privacy Act (FERPA), and Institutional Review Board (IRB) standards. These regulations should be emphasized in professional development. At the foundational Tier 1, learners are taught that privacy laws strictly govern health-related client data; thus, entering a client's protected health information into a third-party AI tool would constitute a breach of confidentiality. Similarly, social work educators and students must remember FERPA protections when using AI in educational settings; student records and academic work should not be shared with AI systems in ways that violate privacy or academic policies. These principles are reinforced at Tier 2 and 3 with more nuance. Intermediate practitioners (Tier 2) learn to apply privacy-by-design principles, such as using only de-identified data with AI tools or opting for platforms that offer encryption and compliance features. Advanced leaders (Tier 3) work to establish organizational protocols that clearly instruct staff on these matters; for example, an agency might approve a HIPAA-compliant AI platform for case documentation while banning unvetted apps from handling any sensitive information (Badillo-Diaz, 2025b).

In addition, all tiers need grounding in the ethics of algorithmic fairness and accountability. This involves recognizing and mitigating biases in AI systems and establishing accountability for the decisions made. Social workers are taught to ask: Who is responsible if an algorithm makes an error that causes harm? At the basic level, it's an awareness that AI decisions can be fallible and biased (Badillo-Diaz, 2025b; Hodgson et al., 2022); at the intermediate level, it's learning techniques to audit an AI's decisions and advocate for clients who may be misclassified; at the advanced level, it's implementing governance mechanisms like an ethics review board for AI or bias mitigation requirements in contracts with AI vendors.

Informed consent is another through-line. The NASW Code of Ethics requires that clients or participants be informed of significant procedures that affect them.

Thus, social workers must strive for transparency with clients regarding the use of AI. At Tier 1, this might mean understanding that if an AI is involved in service delivery, clients should be informed in plain language. At Tier 2, practitioners receive training on how to explain AI tools to clients and obtain consent, or at least assent, for their use. For example, a practitioner might say: “We have a new computer program that helps us identify which services you might need. It will analyze some of your information with your permission and give suggestions, which we will then discuss. You can opt out of this at any time.” Tier 3 leaders will formalize these expectations in policy, possibly requiring that an opt-out clause and robust informed consent procedures accompany any client-facing AI. In the research realm, Tier 3 experts may guide IRBs to update protocols addressing AI, ensuring research participants are informed if AI will be used in the procedures of their research, and ensuring that such use has been reviewed for ethical soundness.

Finally, a cross-cutting theme is the selection of approved, secure AI systems. Especially in practice, social workers should favor tools that meet security and privacy standards. Tier 2 training can include learning to vet an AI tool, verifying whether it’s HIPAA-compliant, checking whether it’s been validated in relevant populations, and determining whether it has a mechanism for human override or review. Tier 3 leaders may develop an official list of sanctioned AI tools for their agency or school, collaborating with IT departments to ensure data protection. They also stay up-to-date with evolving regulations (e.g., new laws on AI or data protection) to update their policies accordingly. The overarching principle is that ethical obligations and legal mandates travel with the use of AI: adopting a new technology does not exempt social workers from their duty to protect client welfare, privacy, and self-determination. By embedding regulatory and ethical training at every tier, the three-tier framework ensures that increased technical skill is always paired with increased ethical responsibility. This prepares social workers not only to use AI effectively but to use it conscientiously, upholding the trust placed in our profession as we navigate the algorithmic age.

5 Exploring the Existing Landscape of AI Training and Framework in Social Work

The conversation about AI in social work is no longer theoretical. In recent years, social work has witnessed a concerted effort to build AI literacy and competencies among professionals through education, training programs, and policy frameworks. Globally, institutions are moving beyond theoretical discussions to implement practical training initiatives that equip social workers with the skills to use AI responsibly. These initiatives span formal higher education curricula, continuing professional education courses, in-agency training programs, and the development of competency frameworks by regulatory and professional bodies.

5.1 Integration of AI Literacy in Social Work Education

Universities and professional schools of social work have begun integrating AI-related content into their curricula and certificate offerings. In the United States, for example, some Master of Social Work (MSW) programs now include modules on technology and data science, while post-graduate certificate programs specifically targeting AI in practice have emerged. New York University's Silver School of Social Work launched a post-master's certificate in "Using Artificial Intelligence to Support Mental Health," reflecting a growing recognition that social workers need specialized training to harness AI tools in clinical and community settings. Similarly, social work educators have called for embedding generative AI content into core curricula. The Council on Social Work Education's technology task force recommended creating a new competency focused on digital literacy, AI ethics, and responsible technology use in future accreditation standards, to ensure graduates are prepared for ethical digital practice and know how to use AI tools responsibly (Council on Social Work Education, 2025). This push in higher education signals a shift from viewing technology as peripheral to recognizing it as central to contemporary social work competency.

Notably, academic initiatives often involve interdisciplinary collaboration. Social work programs are partnering with data science and engineering experts to design training experiences that bridge technical and practice skills. For example, an AI-powered simulation platform called Empathy Helper was developed by a multidisciplinary team of social work educators, mental health professionals, and AI experts at Washington University at St. Louis Brown School of Social Work to enhance student training. Co-led by faculty from a school of social work and a data science professor, this project enables large language models and agentic AI technologies to create realistic virtual clients, allowing students to practice engagement and assessment in a safe, controlled environment. Such collaborations illustrate how integrating expertise from computer science can help social workers build AI competencies within the context of social work values and client-centered practice (Lindsay et al., 2025). Around the world, other universities have followed suit. In Singapore, the National University of Singapore's Department of Social Work offers a two-day executive course on "Artificial Intelligence for Social Work," taught by a data science expert, to rapidly bring practitioners up to speed on AI project lifecycles, machine learning basics, and best practices in AI ethics and governance. It aims to "empower social workers to leverage AI tools ethically and effectively" while maintaining human-centered care (Council on Social Work Education, 2025). Participants learn key AI concepts, the stages of AI projects, and how to identify opportunities for their organizations. These examples from academia underscore a global trend: professional social work education is increasingly embedding AI literacy and AI competencies, often through cross-disciplinary initiatives, to prepare new social workers for a tech-augmented practice environment.

5.2 Continuing Education and Professional Training Programs

Beyond degree programs, a range of continuing education and training programs has emerged worldwide to build AI literacy among practicing social workers. Many of these programs are driven by the urgent need to close the knowledge gap among current practitioners, as studies have found that many social workers still rate their knowledge of AI as low (Council on Social Work Education, 2025). The goal is to enhance AI literacy, enabling social workers to confidently adopt new tools while upholding their ethical standards.

One notable effort is the proliferation of online continuing education courses specifically about AI in social work. For instance, Walden University developed a one-hour asynchronous micro-course in 2024 called “AI Essentials for the Social Worker,” which grants continuing education credit to licensed professionals. This module introduces foundational concepts of generative AI in an accessible manner, aiming to equip practitioners with more informed and confident knowledge about applying AI technologies in social work. As an accredited continuing education course, it aligns with licensure requirements and indicates that AI competency is now an integral part of ongoing professional development.

Open-access and massive open online courses (MOOCs) are also contributing to AI upskilling in the field. A prominent example is the “AI for Social Professionals: Social Work, Psychology, and More” course developed by Vives University College in Belgium. This five-week online course has attracted thousands of enrollees and focuses on leveraging AI responsibly in social services to enhance client care and improve efficiency. The curriculum strikes a balance between practical skills and ethics, where participants explore real-world AI tools for tasks such as client management, documentation, and risk assessment. Through case studies, they learn how to utilize these tools in an ethical and client-centered manner. The popularity of this MOOC suggests a global demand for such knowledge, transcending national boundaries in the social sector.

National professional bodies are likewise providing training resources. In Canada, the Canadian Association of Social Workers hosted webinars as part of an “AI and Social Work Practice” series. One such webinar in March 2025, “Ethical and Responsible Use of AI for Social Work Practice,” was designed to introduce practitioners to the “transformative potential of generative AI” across micro, mezzo, and macro practice, while engaging them in critical discussions on ethics. The session emphasized core social work ethics, including privacy, confidentiality, avoidance of bias, and human-centered care, as nonnegotiables when adopting AI tools. It also included live demonstrations and a hands-on prompt guidebook to help participants develop practical skills in using generative AI in everyday practice settings. This reflects a pedagogical shift toward interactive learning, ensuring that social workers not only learn about AI in theory but also practice using it under guidance, always with an eye to ethical standards. Similarly, in the United States, chapters of the National Association of Social Workers have begun offering continuing education webinars on topics like AI ethics and best practices.

For example, the NASW's Wisconsin Chapter featured a 2024 webinar on "AI and Social Work Ethics" to help practitioners navigate emerging dilemmas, such as informed consent, confidentiality, and documentation, in the age of AI. These professional training efforts indicate that continuing education is rapidly evolving to include AI competency, treating it as essential to modern practice alongside traditional skills.

5.3 Agency Initiatives and Policy Frameworks for AI Competency

At the organizational and policy level, there is growing recognition that social service agencies and regulatory bodies must establish frameworks to guide AI integration and ensure workforce readiness. When agencies adopt AI-driven systems, they face the challenge of training their staff to use these tools appropriately and ethically. Real-world deployments of AI in social work settings have underscored this need. For example, in the United Kingdom, several local authorities began piloting an AI system called "Magic Notes," which automatically transcribes and summarizes case conversations for social workers. The British Association of Social Workers (BASW) has stressed that such tools "must never replace relationship-based practice and decision-making." Crucially, BASW's first practice guidance on AI calls on employers to provide clear training and support for staff using AI, and to designate leadership (e.g., a principal social worker) to oversee the ethical use of AI within agencies (BASW, 2025). The guidance explicitly states that organizations introducing generative AI should ensure practitioners receive appropriate training, clear guidance, and support on the responsible use of these technologies. This reflects a policy-driven framework where professional bodies are not only cautioning about risks but also actively encouraging capacity-building at the agency level. By embedding training expectations into practice standards, bodies like BASW are influencing how agencies roll out AI, essentially mandating that competency development accompanies any AI deployment.

On the ground, several agency-driven training initiatives have started to emerge. Large social service organizations and government departments are developing internal workshops and protocols for their staff when new AI tools are introduced. For example, child welfare agencies that use predictive analytics models for risk assessment have organized training sessions to help social workers understand model outputs and interpret them critically rather than blindly trust algorithmic recommendations. In healthcare social work, hospital systems that deploy AI-based decision support have integrated an AI orientation into their staff development programs, ensuring social workers learn how to co-utilize these tools while still exercising professional judgment. Though many of these internal training courses are not public-facing, they are increasingly guided by formal frameworks and partnerships. A case in point is the collaboration between social service agencies and academic institutions to develop training curricula. Some jurisdictions have partnered with universities to deliver certification programs for employees on topics like

data-informed practice or AI in case management. These partnerships leverage academic expertise and often refer to governance standards to shape content, covering areas such as data privacy law, algorithmic bias, and client consent, which are crucial for agency policy compliance.

Another key aspect of the current landscape is the role of governance and ethical frameworks in shaping the content of training. Professional bodies worldwide are producing guidelines that effectively serve as learning objectives for training programs. The BASW guidance (2025) and NASW's ethical webinars both underscore themes such as bias awareness, transparent use of AI, accountability for decisions, and the primacy of human relationships. These themes are now woven into workshops and courses as standard modules. For example, a continuing education course might include a module on "Ethical Concerns of AI in Social Work," covering privacy, informed consent, and equity issues, echoing the points raised in official standards. Likewise, interdisciplinary bodies have begun issuing broader AI ethics frameworks that social work educators adapt to context-specific training. Social workers are thus being taught not just how to use AI tools but how to evaluate them critically. This capacity for critical appraisal is a competency increasingly highlighted in policy-driven frameworks as essential for the profession's engagement with AI.

In summary, the global landscape reflects a growing, multisectoral effort to develop AI literacy and AI competencies across social work education, training, and practice. Universities are incorporating foundational AI literacy into their curricula, while continuing education programs and professional associations provide applied training to develop role-specific competencies. Agencies and policymakers are beginning to formalize expectations through ethical frameworks and governance tools. Yet, across these efforts, a clear theoretical foundation and cohesive professional guidance remain limited. As a result, many training initiatives operate in silos, underscoring the need for coordinated, evidence-informed strategies that align with the profession's values and standards.

6 From Literacy and Competency to Leadership: A Social Work Professional Agenda for the Algorithmic Age

As this chapter has explored, AI literacy and AI competency form two distinct yet interdependent pillars of social work's digital future. AI literacy provides the knowledge and critical lens, AI competency provides the action orientation and the capacity to implement that knowledge in real-world practice and policy. Together, these two dimensions form a comprehensive framework for professional empowerment in the digital era. Only by maintaining a clear distinction between understanding AI and skillfully using AI can social workers ensure they do not confuse mere familiarity with true capability. The integration of both ensures that social workers not only comprehend the algorithmic tools shaping society but can also harness and guide them in the service of humanity (Badillo-Diaz, 2025a).

The integration of AI literacy and competency positions social workers to confront and mitigate algorithmic injustice. As algorithmic tools become increasingly embedded in child welfare, mental health, benefits eligibility, and criminal justice systems, social workers' dual capacity for ethical discernment and action becomes vital. With these capacities, social workers can critically examine how AI systems may perpetuate or exacerbate structural inequality and take informed steps to prevent or mitigate harm. In practice, this may involve scrutinizing an algorithm's design before implementation, advocating for equity audits and stakeholder consultation, or halting deployment when tools conflict with ethical or legal standards. Far from being passive recipients of innovation, AI-literate and AI-competent social workers act as ethical stewards and co-creators—insisting that technological adoption be guided by relational care, procedural fairness, and community voice (Council on Social Work Education, 2025). In doing so, the profession advances its long-standing commitment to anti-oppressive practice, now extended into the digital domain.

This synthesis of literacy and competency also lays the foundation for a forward-looking professional agenda. To meet the demands of the algorithmic age, these capacities must be intentionally cultivated in education, sustained in practice, and extended through research. Educational programs should embed AI-related knowledge and skills throughout the curriculum. Field placements and continuing education should reinforce this foundation through authentic engagement with AI systems in real-world contexts. Organizations and agencies must invest in workforce development, infrastructure, and participatory design to ensure that ethical use of technology becomes a shared responsibility, not an individual burden. Across all domains, social workers must actively shape research and policy debates by bringing their ethical frameworks and contextual expertise into interdisciplinary conversations on AI governance.

In summary, AI literacy and AI competency are not merely technical supplements to social work; they are essential prerequisites for ethical and effective practice in a rapidly changing world. Their integration enables the profession to respond to rapid technological change not with fear or resignation, but with clarity, confidence, and a deep sense of purpose. This is not simply a matter of adapting to innovation. It is an opportunity for social work to lead: to assert its values, shape the systems that shape lives, and uphold its mission in new and powerful ways in the era of AI.

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References

- Ahn, E., Choi, M., Fowler, P., & Song, I. H. (2025). Artificial intelligence (AI) literacy for social work: Implications for core competencies. *Journal of the Society for Social Work and Research*, 16(1), 9–26.
- Badillo-Diaz, M. (2025a, February 13). Moving beyond AI literacy: Why social workers need AI fluency. — *The AI Social Worker*. <https://www.theaisocialworker.com/blog/building-resilient-communities-the-role-of-ai-in-disaster-response-and-recovery-efforts>

- Badillo-Diaz, M. (2025b, Spring). The impact of AI technology on the social work profession: Benefits, risks, and ethical considerations. *Center for Advanced Studies in Child Welfare* CW360 [Newsletter]. <https://cascw.umn.edu/cw360deg-spring-2025/impact-ai-technology-social-work-profession-benefits-risks-and-ethical>
- Boetto, H. (2025). Artificial intelligence in social work: An EPIC model for practice. *Australian Social Work*, 1–14. <https://doi.org/10.1080/0312407X.2025.2488345>
- British Association of Social Workers [BASW]. (2025, March). *Generative AI and Social Work: Initial Guidance for Practice and Ethics*. BASW. https://basw.co.uk/sites/default/files/2025-03/181372%20Generative%20AI%20and%20Social%20Work%20Practice%20Guidance_0.pdf
- Chan, C., & Holosko, M. J. (2016). A review of information and communication technology enhanced social work interventions. *Research on Social Work Practice*, 26(1), 88–100. <https://doi.org/10.1177/10497315155578>
- Chiu, T. K. F. (2025). AI literacy and competency: Definitions, frameworks, development and future research directions. *Interactive Learning Environments*, 33(5), 3225–3229. <https://doi.org/10.1080/10494820.2025.2514372>
- Chiu, T. K. F., Ahmad, Z., Ismailov, M., & Sanusi, I. T. (2024). What are artificial intelligence literacy and competency? A comprehensive framework to support them. *Computers and Education Open*, 6, 100171. <https://doi.org/10.1016/J.CAEO.2024.100171>
- Council on Social Work Education. (2022). Educational policy and accreditation standards. In *Council on social work education: Washington, DC, USA*. https://www.cswc.org/getmedia/bb5d8afe-7680-42dc-a332-a6e6103f4998/2022-Educational-Policy-and-Accreditation-Standards-%28EPAS%29.pdf?utm_source=chatgpt.com
- Council on Social Work Education. (2025). *CSWE commission on educational policy task force on technology and social work education report*. <https://www.cswc.org/getmedia/3785b60b-5fae-4c39-a0ec-c052edd9f588/COEP-Technology-Report.pdf#:~:text=dimensions%20of%20AI,in%20shaping%20AI%20policies%20that>
- DiSalvo, B., Guzdial, M., Bruckman, A., & McKlin, T. (2014). Saving face while geeking out: Video game testing as a justification for learning computer science. *Journal of the Learning Sciences*, 23(3), 272–315. <https://doi.org/10.1080/10508406.2014.893434>
- Epstein, R. M., & Hundert, E. M. (2002). Defining and assessing professional competence. *JAMA*, 287(2), 226–235. <https://doi.org/10.1001/JAMA.287.2.226>
- Frank, J. R., Snell, L. S., Ten Cate, O., Holmboe, E. S., Carraccio, C., Swing, S. R., Harris, P., Glasgow, N. J., Campbell, C., Dath, D., Harden, R. M., Iobst, W., Long, D. M., Mungroo, R., Richardson, D. L., Sherbino, J., Silver, I., Taber, S., Talbot, M., & Harris, K. A. (2010). Competency-based medical education: Theory to practice. *Medical Teacher*, 32(8), 638–645. <https://doi.org/10.3109/0142159X.2010.501190>
- Freire, P. (2007). Pedagogy of the oppressed. In P. Kupperts & G. Robertson (Eds.), *The community performance reader* (pp. 24–27). Routledge. <https://doi.org/10.4324/9781003060635-5>
- Goldkind, L. (2021). Social work and artificial intelligence: Into the matrix. *Social Work*, 66(4), 372–374. <https://doi.org/10.1093/sw/swab028>
- Hodgson, D., Goldingay, S., Boddy, J., Nipperess, S., & Watts, L. (2022). Problematising Artificial Intelligence in social work education: Challenges, issues and possibilities. *British Journal of Social Work*, 52(4), 1878–1895. <https://doi.org/10.1093/BJSW/BCAB168>
- Laugksch, R. C. (2000). Scientific literacy: A conceptual overview. *Science Education*, 84(1), 71–94.
- Lindsay, R., Yang, Y., & An, R. (2025). *A pilot study of empathy helper, a generative Artificial Intelligence platform for communication practice in graduate social work*. [Manuscript submitted for publication, under peer review].
- Liu, X., Zhang, L., & Wei, X. (2025). Generative artificial intelligence literacy: Scale development and its effect on job performance. *Behavioral Sciences*, 15(6), 811. <https://doi.org/10.3390/BS15060811>
- Long, D., & Magerko, B. (2020, April 21). What is AI literacy? Competencies and design considerations. *Conference on human factors in computing systems – Proceedings*. <https://doi.org/10.1145/3313831.3376727>

- Mikalef, P., Islam, N., Parida, V., Singh, H., & Altwaijry, N. (2023). Artificial intelligence (AI) competencies for organizational performance: A B2B marketing capabilities perspective. *Journal of Business Research*, 164, 113998. <https://doi.org/10.1016/J.JBUSRES.2023.113998>
- Miller, G. E. (1990a). The assessment of clinical skills/competence/performance. *Academic Medicine: Journal of the Association of American Medical Colleges*, 65(9 Suppl), S63–S67. <https://doi.org/10.1097/00001888-199009000-00045>
- Miller, G. E. (1990b). The assessment of clinical skills/competence/performance. *Academic Medicine*, 65(9), S63–S67. <https://doi.org/10.1097/00001888-199009000-00045>
- National Association of Social Workers [NASW]. (2021). *NASW code of ethics*. <https://www.socialworkers.org/About/Ethics/Code-of-Ethics/Code-of-Ethics-English>
- Nordesjö, K., Scaramuzzino, G., & Ulmestig, R. (2022). The social worker-client relationship in the digital era: A configurative literature review. *European Journal of Social Work*, 25(2), 303–315. <https://doi.org/10.1080/13691457.2021.1964445>
- Nuwasiima, M., Ahonon, M. P., & Kadiri, C. (2024). The role of Artificial Intelligence (AI) and machine learning in social work practice. *World Journal of Advanced Research and Reviews*, 24(1), 080–097. <https://doi.org/10.30574/WJARR.2024.24.1.2998>
- Perchik, J. D., Smith, A. D., Elkassem, A. A., Park, J. M., Rothenberg, S. A., Tanwar, M., Yi, P. H., Sturdivant, A., Tridandapani, S., & Sotoudeh, H. (2023). Artificial Intelligence literacy: Developing a multi-institutional infrastructure for AI education. *Academic Radiology*, 30(7), 1472–1480. <https://doi.org/10.1016/J.ACRA.2022.10.002>
- Perron, B. E., Taylor, H. O., Glass, J. E., & Margerum-Leys, J. (2010). Information and communication technologies in social work. *Advances in Social Work*, 11(2), 67. <https://doi.org/10.18060/241>
- Reamer, F. G. (2013). Social work in a digital age: Ethical and risk management challenges. *Social Work*, 58(2), 163–172. <https://doi.org/10.1093/SW/SWT003>
- Reamer, F. G. (2023). Artificial Intelligence in social work: Emerging ethical issues. *International Journal of Social Work Values and Ethics*, 20(2), 52–71. <https://doi.org/10.55521/10-020-205>
- Rodriguez, M. Y., Goldkind, L., Victor, B. G., Hiltz, B., & Perron, B. E. (2024). Introducing generative artificial intelligence into the MSW curriculum: A proposal for the 2029 educational policy and accreditation standards. *Journal of Social Work Education*, 60(2), 174–182.
- Tsz, D., Ng, K., Ka, J., Leung, L., Kai, S., Chu, W., & Shen Qiao, M. (2021). NC-ND license conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, 100041. <https://doi.org/10.1016/j.caeai.2021.100041>
- UNESCO. (2024). AI competency framework for students. In *AI competency framework for students*. UNESCO. <https://doi.org/10.54675/JKJB9835>
- UNESCO. (2025, February 20). *What you need to know about UNESCO's new AI competency frameworks for students and teachers*. UNESCO News. <https://www.unesco.org/en/articles/what-you-need-know-about-unescos-new-ai-competency-frameworks-students-and-teachers>

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